

Lab Problem 9

Objective: Implement kernel functions and explore the kernel trick using the XOR dataset to demonstrate why kernels are needed for non linearly separable data.

Dataset: Generate the **XOR Dataset** by sampling 100 points from four 2D Gaussian distributions centered at **(+1,+1)**, **(-1,-1)**, **(+1,-1)**, and **(-1,+1)** with unit variance. Assign class label **+1** to the first two quadrants and **-1** to the remaining two. Use `random_state = 42`

Task 1 — Visualize the XOR Dataset

Plot the XOR dataset as a scatter plot with the two classes in distinct colors. Label the axes as x_1 and x_2 , include a legend for Class +1 and Class -1.

Task 2 — Implement Kernel Functions

Implement the following three kernel functions from scratch using NumPy only. **No library kernel calls are permitted.**

- **Linear Kernel:** $K(x, z) = x^T z$
- **Polynomial Kernel:** $K(x, z) = (x^T z + 1)^2$ with degree = 2
- **RBF Kernel:** $K(x, z) = \exp(-\gamma \|x - z\|^2)$ with $\gamma = 0.5$

Task 3 — Compute and Print Kernel Matrices

Using the **first 8 points** of the dataset, compute the 8×8 kernel matrix for each of the three kernels. Print each matrix rounded to 2 decimal places with an appropriate label.

Task 4 — Kernel Matrix Heatmaps

Plot the three kernel matrices from Task 3 as heatmaps in a single figure with 1 row and 3 columns. Use `cmap = "coolwarm"`, add a colorbar to each subplot, and title each subplot with its kernel name.

Task 5 — Feature Map Visualization

For the degree 2 polynomial kernel, the explicit feature map is given by:

$$\Phi(x) = [x_1^2, \sqrt{2} \cdot x_1 x_2, x_2^2]$$

Transform the full XOR dataset using this feature map and plot the transformed points in a **3D scatter plot**. Color the points by class label, label the axes as ϕ_1 , ϕ_2 , ϕ_3 .